

end equipment
Summary
technology
Samples
Production

System

MCU core

DSP core

SDRAM I/F

Ext. Mem. I/F

Ext. Host CPU I/F

SDRAM bus sharing

Boot Modes

Clock Control

ARM Internal RAM / ROM

ARM cache memory

Peripherals

Timers

Watch Dog Timer

I2C Single Master

VLINQ (high-speed port)

UART

Serial Port

GPIO

Interrupt Controller

I/O Peripherals

IrDA

USB Device

USB Host

CompactFlash

SmartMedia

MultiMediaCard/SD

Memory Stick

DM310	
ARM9 + C54x + iMX8 + VLCD	
0.13u CMOS	
now	
3Q03	
DM310 : System	
ARM925	144* MHz
C5409A	72* MHz
iMX/VLCD	144* MHz
flexible priority	85* MHz
16/32-bit I/F	max: 128MB
1 shared chip-select	
8/16-bit I/F	max: 8MB
x	
x	
- Linear Flash	
individual module disable plus powerdown	
PLL powerdown	
8 KB	-
16 KB I-cache	8 KB D-cache
DM310 : Peripherals	
4	
Timers can be used to generate accurate pulse widths on GPIO	
✓	
x	
x	
2	
2	w/ DMA
32	
8 General interrupts Any GPIO can be configured as interrupt - each general interrupt can be mapped to 1 of 4 GPIOs	
29 interrupt channels	
DM310 : I/O Peripherals	
x	
v 1.1	w/DMA 8/16-bit access
✓	w/ DMA*
✓	w/ DMA
✓	w/ DMA
✓	w/ DMA
✓	w/ DMA

DM270	
DSC, Wireless, DVC, PDA	
ARM7 + C54x + iMX4 + VLCD	
0.13u CMOS	
now	
3Q03	
DM270 : System	
ARM7TDMI	80 MHz
C5409A	100 MHz
iMX/VLCD	180* MHz
flexible priority	100 MHz
16/32-bit I/F	max: 64MB
flexible, w/ 2 user chip-selects	
8/16/32-bit I/F	max: 8-32MB
✓	
w/Linear Flash, CFC/SSFDC	
- Linear Flash	
- SIF / External Host	
individual module disable plus powerdown	
PLL and Oscillator powerdown	
32 KB	-
8 KB I-cache	-
DM270 : Peripherals	
4	
✓	
✓	
x	
2	
2	w/ DMA
34	
GIO 0:7 can be external interrupts	
39 interrupt channels	
mapping for each channel to FIQ/IRQ hardware configurable int priority hardware int entry table	
DM270 : I/O Peripherals	
x	
v 1.1	w/ DMA and 8/16 bit access
x	
✓	w/ DMA
✓	w/ DMA
✓	w/ DMA
✓	w/ DMA

DM320	
DSC, Wireless, DVC, PDA, JB	
ARM9 + C54x + iMX8 + VLCD + DCT + SEQ	
0.13u CMOS	
Jan-04	
3Q04	
DM320 : System	
ARM926EJS	160* Mhz
C5409A	100* MHz
iMX/VLCD/DCT	175* MHz
flexible priority	100* MHz
16/32-bit I/F	max: 256MB
flexible, w/ 2 user chip-selects	
8/16/32-bit I/F	max: 8-32MB
✓	
w/Linear Flash, CFC/SSFDC	
- Linear Flash	
- SIF / External Host	
- NAND-flash (via ARM ROM)	
individual module disable plus powerdown	
PLL and Oscillator powerdown	
16 KB	8 KB
16 KB I-cache	8 KB D-cache
DM320 : Peripherals	
4	
Timers can be used to generate accurate pulse widths on GPIO	
✓	
✓	
✓	
Supports VLINQ devices, including TI V2USB w/ High-speed slave port and full-speed host port	
2	
2	w/ DMA
~34	
GIO 0 is wakeup signal GIO 1:15 can be external interrupts	
~39 interrupt channels	
mapping for each channel to FIQ/IRQ hardware configurable int priority hardware int entry table	
DM320 : I/O Peripherals	
x	
OTG	w/ DMA
✓	w/ DMA
✓	w/ DMA
✓	w/ DMA
✓	w/ DMA

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Imaging Peripherals

Digital Video Inputs
CCD Controller

- VGA 30 fps Sensors

Preview Engine

Hardware 3A Statistics

On-Screen Display

- Video Windows

- Bitmap Windows

- Cursor Windows

- Color LUT

- Bitmap Window

Enhancements

Video Encoder

- DACs

- Digital video out I/F

Digital LCD

RGB 8-bit/pixel

YUV 8-bit (ITU-R BT.656)

YUV 16-bit (ITU-R BT.601)

LCD TG (Epson, Casio)

Microdisplay (Displaytech)

RGB666 LCD support

STN LCD support

DSP Subsystem

Internal Ram

Expanded Ram

ROM

iMX Coprocessor

Image Buffers

Bitstream Buffer

Command Buffer

Coefficient Buffer

DCT/IDCT

Quantization

Inverse Quantization

Variable Length Encode

Variable Length Decode

Micro-Sequencer controller

Image Buffer SDRAM DMA

DSP/HPI SDRAM DMA

McBSP

DSP GIO

DM310	
ARM9 + C54x + iMX8 + VLCD	
0.13u CMOS	
now	
3Q03	
DM310 : Imaging Peripherals	
BT.601 IF (8-bit)	
12-bits	40 MHz
x	
✓	improved image processing, SDRAM input, image resizing
x	
✓	
2	
2	
2	1 bitmap
256 color	ROM or RAM
✓	
3	Comp, Y/C, or R,G,B + powerdown ctl
✓	8-bit RGB
✓	
✓	
x	
x	
x	
x	
x	
x	
x	
DM310 : DSP Subsystem	
32kW	
32kW (Program Only)	
16kW	
8 MACs	1152
2 x 8KW	
1 x 2KW	
8 KW, unified	
16 bits	
via iMX	
QIQ	
VLCD	JPEG, MPEG1, MPEG4, H.263, MPEG2
1 Word unit	2d in SDRAM and Image Buffer
2	CLKS input
15	shared w/ARM

DM270	
DSC, Wireless, DVC, PDA	
ARM7 + C54x + iMX4 + VLCD	
0.13u CMOS	
now	
3Q03	
DM270 : Imaging Peripherals	
CCIR Rec. 601/656 IF	
14-bits	40 MHz
x	
✓	improved image processing, SDRAM input, image resizing
✓	
✓	
2	
2	
2	1 bitmap
256 color	ROM or RAM
✓	
1	Comp + powerdown ctl
✓	8-bit RGB CIR Rec.656 output (YUV)
✓	
✓	
✓	
✓	
LDM-0311-DN	
✓	
DM270 : DSP Subsystem	
32kW	
32kW	
16kW	
4 MACs	720 MIPS
2 x 8KW	
1 x 2KW	
2 KW	
3.5 KW	16 bits
iMX	
QIQ (in VLCD)	
VLCD	JPEG, MPEG1, MPEG4, H.263, MPEG2
1 Word unit	2d in SDRAM and Image Buffer
2	CLKS input
0	

DM320	
DSC, Wireless, DVC, PDA, JB	
ARM9 + C54x + iMX8 + VLCD + DCT + SEQ	
0.13u CMOS	
Jan-04	
3Q04	
DM320 : Imaging Peripherals	
CCIR Rec. 601/656 IF	
14-bits	40 MHz
Flexible front end to reformat data from VGA 30 fps sensors to standard Bayer format	
✓	improved image processing, SDRAM input, image resizing
✓	
✓	
2	
2	
2	1 bitmap
256 color	ROM or RAM
- 16-bit RGB mode (64K colors)	
- NTSC/PAL square pixel scaling	
✓	
1	Comp + powerdown ctl
✓	8-bit RGB CIR Rec.656 output (YUV)
✓	
✓	
✓	
QDM-0076-MV5	
LDM-0311-DN	
✓	
✓	
DM320 : DSP Subsystem	
32kW	
32kW	
16kW	
8 MACs	1400 MIPS
3 x 4KW	
4 KW	
6 KW	16 bits
DCT/IDCT hardware	
QIQ (in VLCD)	
VLCD	JPEG, MPEG1, MPEG4, H.263, MPEG2
✓	2KW Pgm Mem
1 Word unit	2d in SDRAM and Image Buffer
✓	
2	CLKS input
0	